SLOW MOTION OF GRADIENT FLOWS FROM BAD INITIALISATION

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Abstract. In recent years, a phenomenon has been documented in the training of machine learning models in which i) training loss quickly tends towards its minimum value, while ii) the test loss takes much longer to decay towards its minimal value. Interpreting this as an optimisation mismatch between convergence of function values and algorithm iterates, we demonstrate a simple analysis of gradient flow which reproduces some of this behaviour.

1. Statement and Proof

Theorem 1. *Let $F : \mathbf{R}^d \to \mathbf{R}$ be smooth and bounded, such that the gradient flow $\dot{x} = -\nabla F(x)$ has a global solution. Write $x_\star$ for the unique minimiser of f, with $F_\star = F(x_\star)$, and let x_0 be such that $F(x_0) > F_\star$.*

(1) *For times $0 \leq t \leq \|x_0 - x_\star\|^2 / (F(x_0) - F_\star)$, it holds that*

$$\|x_t - x_\star\| \geq \|x_0 - x_\star\| - \sqrt{t \cdot \{F(x_0) - F_\star\}}.$$

(2) *In particular, if for some $\varepsilon > 0$, $r > 0$, it holds that*

$$F(x_0) - F_\star \leq \varepsilon, \qquad \|x_0 - x_\star\| \geq r,$$

then for any $\vartheta \in (0,1)$, it holds that

$$0 \leq t \leq \vartheta^2 \cdot \frac{r^2}{\varepsilon} \implies \|x_t - x_\star\| \geq (1-\vartheta) \cdot r.$$

That is, if at initialisation, F is close to its infimum, but x is far from the minimiser, then for times of order r^2/ε, x_t remains far from the minimiser, up to a constant multiplicative factor.

Proof. Under the gradient flow, $\frac{\mathrm{d}}{\mathrm{d}t} F(x_t) = -\|\nabla F(x_t)\|^2$. Integration then shows that

$$\int_{0 \leq s \leq t} \|\dot{x}_s\|^2 \, \mathrm{d}s = F(x_0) - F(x_t) \leq F(x_0) - F_\star.$$

On the other hand, Cauchy-Schwarz shows that

$$\|x_t - x_0\|^2 = \left\| \int_{0 \leq s \leq t} \dot{x}_s \, \mathrm{d}s \right\|^2 \leq t \cdot \int_{0 \leq s \leq t} \|\dot{x}_s\|^2 \, \mathrm{d}s \leq t \cdot \{F(x_0) - F_\star\}.$$

By the triangle inequality, one has

$$\begin{aligned} \|x_t - x_\star\| &\geq \|x_0 - x_\star\| - \|x_t - x_0\| \\ &\geq \|x_0 - x_\star\| - \sqrt{t \cdot \{F(x_0) - F_\star\}}, \end{aligned}$$

and the first claim follows. The second claim follows by algebraic manipulation. □

2. Discussion

For this result to tell the intended qualitative story, it must be the case that $r^2/\varepsilon \gg 1$, i.e. that the initial x_0 satisfies

$$F(x_0) - F_\star \ll \|x_0 - x_\star\|^2 .$$

Translated into modern optimisation terminology, this represents a failure of the 'quadratic growth' condition for F at $x = x_0$ (at least, failure to have a good constant on the domain of interest); this then implies failure of the stronger Polyak-Łojasiewicz condition, and the yet stronger condition of strong convexity. With this being said, it is really a rather finitary condition on the initialisation of the gradient flow: if the initial loss is small, but the distance to the minimiser is large, then this distance will remain large for a while.

We note in passing that the uniqueness of minimisers for F is also not essential for this argument, and one can equally replace 'far from the minimiser' to 'far from the set of minimisers' without affecting the conclusions. Moreover, the result can even be localised to show slow escape of the gradient flow from neighbourhoods of metastable configurations. That is, let U be an open set containing x_0 such that for some $\varepsilon > 0$,

$$x \in \bar{U} \implies F(x) \geq F(x_0) - \varepsilon,$$

and write $r = \mathsf{dist}(x_0, \partial U)$, $\tau_U = \inf\{t > 0 : x_t \notin U\}$. It then holds that $\tau_U \geq r^2/\varepsilon$, much as before. Arguments of this type play an important role in the energy-based analysis of metastability and slow motion for gradient flows, notably in the works of Bronsard–Kohn and Otto–Reznikoff on the Allen–Cahn equation.

As a final remark, the estimate is sharp, even in dimension one, as can be seen by considering gradient flow for a linear objective on an interval. Moreover, if F is everywhere at least $F(x_0)$ on ∂U, then escape is impossible altogether, by monotonicity of the objective. The estimate can hence be viewed as quantifying the transition between complete trapping by an energy barrier and slow motion when only a small amount of energy can be dissipated. Finally, the argument relies only on energy dissipation and the Cauchy–Schwarz inequality, and consequently extends immediately to Hilbert space and suitable metric space gradient flows.